SIMPLE SYSTEM OVERVIEW - TWO DOOR GROUPS

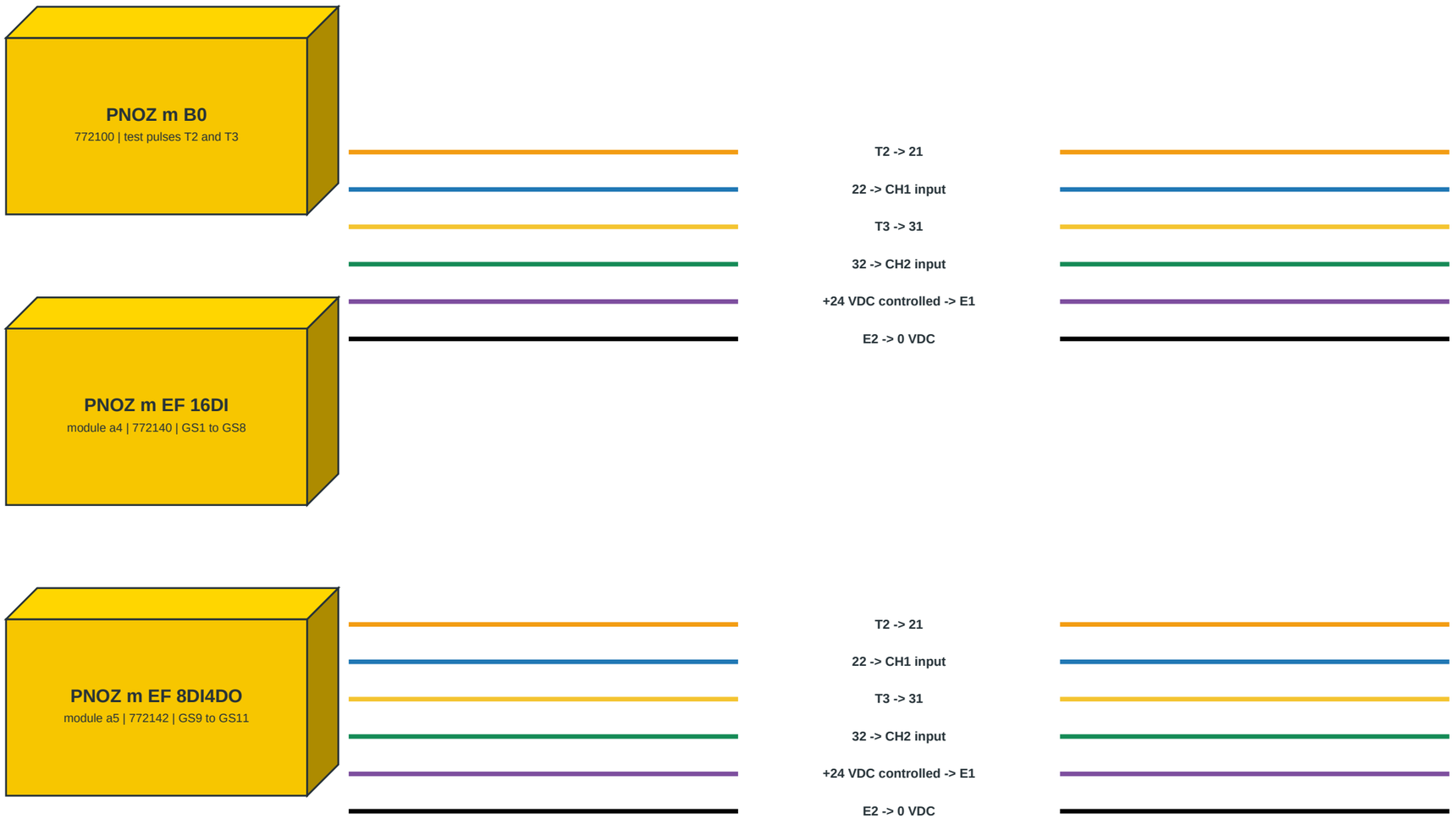
Start here. Pages 4 and 5 then show the exact terminals and every individual conductor.

PNOZ m B0
772100 | test pulses T2 and T3

PNOZ m EF 16DI
module a4 | 772140 | GS1 to GS8

PNOZ m EF 8DI4DO
module a5 | 772142 | GS9 to GS11

24 VDC / 0 VDC
through F-ULx and KGSx to E1/E2



DOOR GROUP A

PSEN me1S/1AR
article 570001

21 22 31 32 E1 E2

GS1-GS8 | inputs a4.i0 to a4.i15

DOOR GROUP B

PSEN me1S/1AR
article 570001

21 22 31 32 E1 E2

GS9-GS11 | inputs a5.i0 to a5.i5

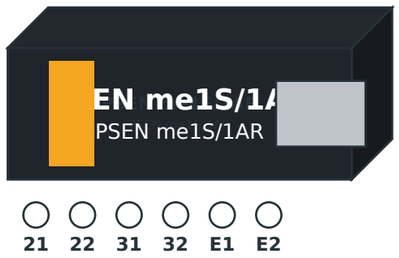
HOW TO READ THIS DRAWING

1. B0 supplies T2 and T3. 2. a4 reads GS1-GS8. 3. a5 reads GS9-GS11. 4. Controlled +24 VDC unlocks E1-E2.

For the exact input pair of every door, use page 3 or page 5. For the exact six-wire connection, use page 4.

ONE DOOR - DETAILED SENSOR AND 8-CORE CABLE

Repeat this circuit for GS1 through GS11; only the assigned CH1 and CH2 PNOZ inputs change.

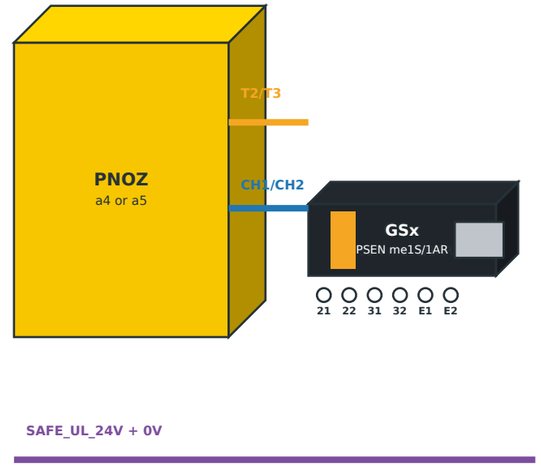


Sensor terminals

21-22	NC guard-position safety contact
31-32	NC lock-position safety contact
E1-E2	24 V unlock solenoid
13-14	Auxiliary NO contact (optional)
43-44	Auxiliary NO contact (optional)

Recommended cable: 8 conductors per door

Core 1: T2	-> PSEN 21
Core 2: PSEN 22	-> PNOZ CH1 input
Core 3: T3	-> PSEN 31
Core 4: PSEN 32	-> PNOZ CH2 input
Core 5: SAFE_UL_24V via F-ULx/KGSx	-> PSEN E1
Core 6: PSEN E2	-> 0 V DC
Core 7: Spare / optional auxiliary	-> 13-14 or 43-44
Core 8: Spare / optional return	-> non-safety input/common



Cable count rule

Minimum functional: 6 conductors

Recommended: 8 conductors (2 spare/aux)

Both auxiliary contacts used: 10 conductors

Cable cross-section, wire numbers, fuses and cabinet terminal blocks must be released by the electrical design; they are intentionally not invented here.

COMPLETE DOOR SCHEDULE AND INSTALLATION TOTALS

Exact confirmed input assignment from the PNOZmulti project screenshots supplied by the operator.

Door	Location	PSEN safety path	CH1 Pilz input	CH2 Pilz input	Cable	Unlock branch
GS1	FRONT 1	T2-21/22 + T3-31/32	a4.i0	a4.i1	8-core	F-ULx / KGSx -> E1-E2
GS2	FRONT 2	T2-21/22 + T3-31/32	a4.i2	a4.i3	8-core	F-ULx / KGSx -> E1-E2
GS3	FRONT 3	T2-21/22 + T3-31/32	a4.i4	a4.i5	8-core	F-ULx / KGSx -> E1-E2
GS4	FRONT 4	T2-21/22 + T3-31/32	a4.i6	a4.i7	8-core	F-ULx / KGSx -> E1-E2
GS5	REAR 1	T2-21/22 + T3-31/32	a4.i8	a4.i9	8-core	F-ULx / KGSx -> E1-E2
GS6	REAR 2	T2-21/22 + T3-31/32	a4.i10	a4.i11	8-core	F-ULx / KGSx -> E1-E2
GS7	REAR 3	T2-21/22 + T3-31/32	a4.i12	a4.i13	8-core	F-ULx / KGSx -> E1-E2
GS8	REAR 4	T2-21/22 + T3-31/32	a4.i14	a4.i15	8-core	F-ULx / KGSx -> E1-E2
GS9	RIGHT 1	T2-21/22 + T3-31/32	a5.i0	a5.i1	8-core	F-ULx / KGSx -> E1-E2
GS10	RIGHT 2	T2-21/22 + T3-31/32	a5.i2	a5.i3	8-core	F-ULx / KGSx -> E1-E2
GS11	LEFT 1	T2-21/22 + T3-31/32	a5.i4	a5.i5	8-core	F-ULx / KGSx -> E1-E2

11 x 8-core

door cables

88

installed conductors

66

essential conductors used

22

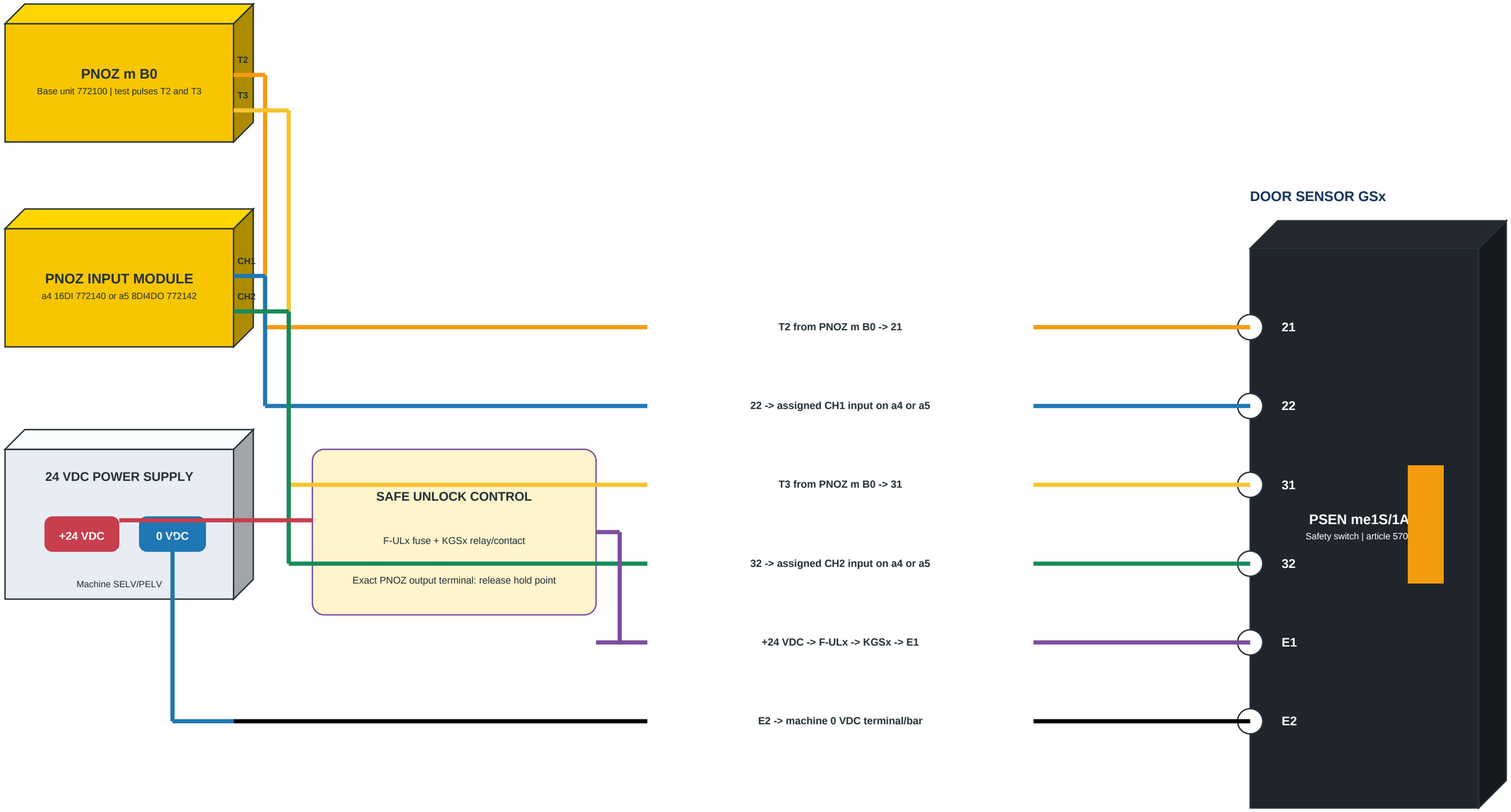
spare / optional auxiliary

Commissioning check: verify each physical door identity, both channel transitions, cross-short diagnostics, lock state, controlled unlock and fault reset.

Do not energize or bypass the safety function from this concept drawing. Use the released schematic and validated PNOZ project for final commissioning.

ONE DOOR - EXACT DEVICES, TERMINALS, 24 VDC AND 0 VDC

Use this large circuit as the wiring pattern for every GS door. The input numbers change according to page 5.



+24 VDC goes through F-ULx and KGSx before E1.

E2 returns directly to the machine 0 VDC terminal/bar, not to a PNOZ safety input.

ALL 11 DOORS - INDIVIDUAL COLORED CONDUCTORS

Exact sensor and Pilz names are shown. Red/blue DC rails are separate from the four safety-channel conductors.

PNOZ m B0
772100 - T2/T3

PNOZ m EF 16DI
a4 - 772140

PNOZ m EF 8DI4DO
a5 - 772142

24 VDC / 0 VDC
through F-ULx + KGSx

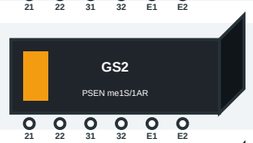
GS1 - FRONT 1 | PSEN me1S/1AR 570001 | PNOZ m EF 16DI a4

T2 -> 21
22 -> a4.i0
T3 -> 31
32 -> a4.i1
+24V controlled -> E1
E2 -> 0V



GS2 - FRONT 2 | PSEN me1S/1AR 570001 | PNOZ m EF 16DI a4

T2 -> 21
22 -> a4.i2
T3 -> 31
32 -> a4.i3
+24V controlled -> E1
E2 -> 0V



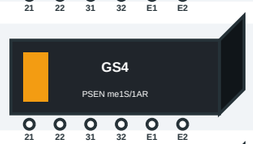
GS3 - FRONT 3 | PSEN me1S/1AR 570001 | PNOZ m EF 16DI a4

T2 -> 21
22 -> a4.i4
T3 -> 31
32 -> a4.i5
+24V controlled -> E1
E2 -> 0V



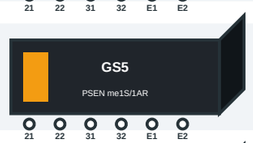
GS4 - FRONT 4 | PSEN me1S/1AR 570001 | PNOZ m EF 16DI a4

T2 -> 21
22 -> a4.i6
T3 -> 31
32 -> a4.i7
+24V controlled -> E1
E2 -> 0V



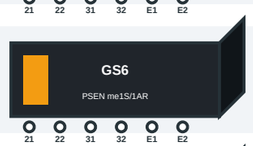
GS5 - REAR 1 | PSEN me1S/1AR 570001 | PNOZ m EF 16DI a4

T2 -> 21
22 -> a4.i8
T3 -> 31
32 -> a4.i9
+24V controlled -> E1
E2 -> 0V



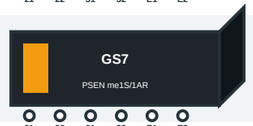
GS6 - REAR 2 | PSEN me1S/1AR 570001 | PNOZ m EF 16DI a4

T2 -> 21
22 -> a4.i10
T3 -> 31
32 -> a4.i11
+24V controlled -> E1
E2 -> 0V



GS7 - REAR 3 | PSEN me1S/1AR 570001 | PNOZ m EF 16DI a4

T2 -> 21
22 -> a4.i12
T3 -> 31
32 -> a4.i13
+24V controlled -> E1
E2 -> 0V



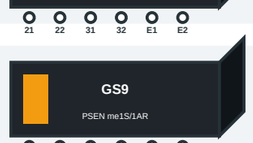
GS8 - REAR 4 | PSEN me1S/1AR 570001 | PNOZ m EF 16DI a4

T2 -> 21
22 -> a4.i14
T3 -> 31
32 -> a4.i15
+24V controlled -> E1
E2 -> 0V



GS9 - RIGHT 1 | PSEN me1S/1AR 570001 | PNOZ m EF 8DI4DO a5

T2 -> 21
22 -> a5.i0
T3 -> 31
32 -> a5.i1
+24V controlled -> E1
E2 -> 0V



GS10 - RIGHT 2 | PSEN me1S/1AR 570001 | PNOZ m EF 8DI4DO a5

T2 -> 21
22 -> a5.i2
T3 -> 31
32 -> a5.i3
+24V controlled -> E1
E2 -> 0V



GS11 - LEFT 1 | PSEN me1S/1AR 570001 | PNOZ m EF 8DI4DO a5

T2 -> 21
22 -> a5.i4
T3 -> 31
32 -> a5.i5
+24V controlled -> E1
E2 -> 0V



T2 -> 21

22 -> CH1

T3 -> 31

32 -> CH2

+24 VDC controlled -> E1

E2 -> 0 VDC